



Midterm Exam Phy-111C

Time of exam: 60 minutes

24

الاسم بالعربية : _____ : مرس المادة : _____ : موعد المختبر : _____

Note: When you need consider the acceleration of gravity $g = 10 \text{ m/s}^2$

Q(1)	1	2	3	4	5	6	7	8	9	10	Q(2)	1	2	3	4	5
	B	B	D	C	D	A	D	D	B	B		X	X	F	T	F

Q.1: Circle the correct answers, and then move these answers to the table : (20 marks)

1) The instrument which used to take this reading (2.265cm) is:

A) ruler B) micrometer C) Vernier caliper D) none of all

2) For an (3) forces equilibrium system as shown in the table,

The magnitude of (F_3)

A) 0.99 N B) 1.06 N C) 0.92 N D) 9. N

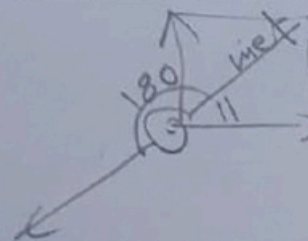
3) The direction of (F_3) :

A) 229.4° B) 49.4° C) 221.2° D) 41.2°

mass	force	direction	x-cm	y-cm
80 g	F1	?	0.8	0
70 g	F2	?	0	0.7
M3	F3	?	?	?

4) Using significant figures rules $(8.58 \times 3.6) =$

A) 30.9 B) 30.6 C) 31. D) 30.60



In Newton's second law experiment, if $m_1=0.35$ and $m_2=10g$.
(kg)

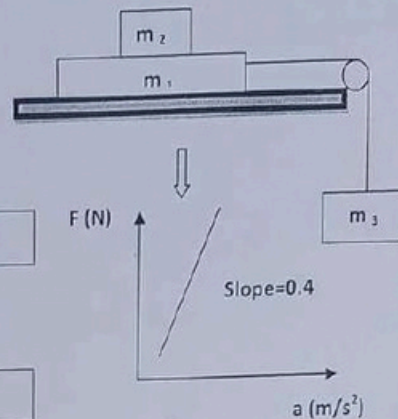
Answer question (5) and (6)

5) The total force which moves these masses is:

A) 0.1 N	B) 0.35 N	C) 0.3 N	D) 0.4 N
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6) The magnitude of (m_3) in kg, is:

A) 0.04	B) 0.03	C) 0.035	D) 0.01
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In uniform circular motion experiment, if you got the graph shown,

Answer question (7) and (8)

7) The mass which is rotate in uniform circular motion (m) in (kg) is:

A) 0.14	B) 0.014	C) 0.1	D) 0.2
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8) When $a_c=295.78 \text{ m/s}^2$, and ($r=30\text{cm}$). the time of (20) revolutions is:

A) 6.67 s	B) 8.67 s	C) 9.2 s	D) 4 s
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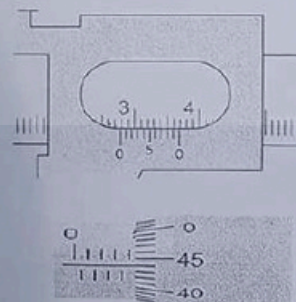
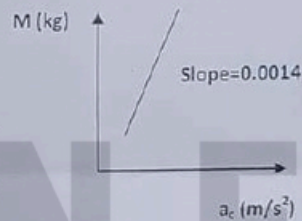
9) Using significant figures rules, ($74025000 \div 987$) is:

A) 0.8×10^5	B) 0.750×10^5	C) 0.75×10^5	D) 75000
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10) The sum of micrometer and caliper reading in (mm) is:

(using (s.f) rules)

A) 32.24	B) 32.2	C) 33.	D) 33.1
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Q (2): Put (T) in front of the correct phrase, and (F) In front of the wrong phrase. Then move these answers to the table. (10marks)

- 1) The relation between (a) and (1/m), when the force is constant, is linear directly proportional. (~~F~~). ~~F~~ T
- 2) The benefit of crocodile clip in uniform circular motion is to keep the radius constant when we rotate the small mass (m). (T).
- 3) In Newton's second law experiment, you can calculate always (V_f), as $V_f = 1/t^2$. (F).
- 4) In Newton's second law experiment, the decline of air track acts on results. (T).
- 5) The uncertainty for the area of circle, is $\Delta A = (2\Delta r/r)A$. (T).

$$\pi r^2 \quad \Delta A = \frac{2\Delta r}{r} A$$

Good luck for all

$$Mg = \frac{v^2}{r}$$

$$4\pi r$$

$$0,4 = 3,26 + m_2$$

$$m_2 g =$$

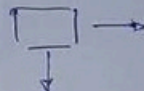
$$\downarrow \textcircled{a} \propto \frac{1}{m} \uparrow$$

$$Mg = 4\pi^2 r m f^2$$

$$\frac{Mg}{10} = \frac{14}{10}$$

$$0,01$$

$$0,36$$



$$\frac{10}{1000}$$

$$1,4$$

$$4 \times$$

$$m a_c = 4\pi^2 r m f^2$$

$$m_2 g - T = 3,6$$

$$m a_c = \frac{1,6}{0,01}$$

$$295,78 = \frac{11,8}{0,01}$$

$$T =$$

$$5,6$$

$$\Sigma F_y = T$$

$$295,78$$

$$0,104$$

$$\Sigma F_y = 3,6$$

$$27,7$$

$$102$$

$$T - m_2 g = 3,6$$

$$4,44 \text{ mm}$$



Midterm Exam Phy-111(A)

الاسم بالعربية :	مدرس المادة :	عدد المسجلين :
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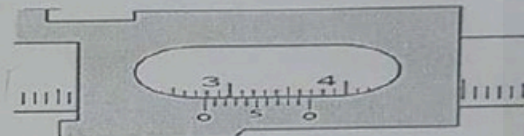
• Q 1: Fill the blank with suitable answer :

(13.5 marks)

1+7

- 1) If we turned a closed micrometer (5) turns and (1/5) of a turn, then the micrometer reading is (~~2.88~~ mm).

$$5.20 \text{ mm} \\ = 2.6 \text{ mm}$$



- 2) $L = (\text{2.88} \text{ mm}) \pm \Delta L = (0.005 \text{ mm})$ 0.05 mm

$$2.88 \pm 0.005 \text{ mm}$$



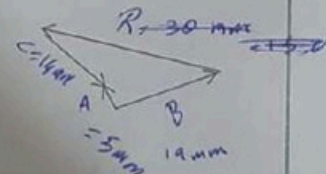
- 3) If the diameter for a ball was measured using a micrometer as shown then: $\Delta A = (\pm 0.16 \text{ mm}^2)$ 0.133 mm^2

- 4) If the scale used to draw this vector diagram for three forces

was (2mm = 1 N), then the magnitude of the net force is (12 N)

$$R = 4 \text{ mm} \rightarrow 2 \text{ N}$$

$$2 \text{ mm} = 1 \text{ N} \\ 4 \text{ mm} = 2 \text{ N} \\ 2 \times 4 = 8 \text{ N} \times 2 = 16 \text{ N}$$



- 5) The instrument used to take this reading (13.3 cm) is (Vernier caliper, Micrometer, Ruler).

(Ruler).

6) Using significant rules to find $306000/45.0 = (680 \times 10^1)$ ✓).

7) In gravitational potential energy experiment, before start move toward down from ($h_i=100\text{cm}$) the mass was ($m=20\text{g}$) and ($M=200\text{g}$), then the kinetic energy when (m) reach ($h=40\text{cm}$) is: (5.92 joule) ✓

8) For the above question the final velocity (V_f) when m reach ground is: (✓) m/s).

9) Using significant rules to find $4.207+3.78 = (7.99)$ ✓

• Q2 : Circle the correct answer: (10.5 marks)

1) If a student used a ruler with a maximum range (15 cm) to measure the length of a table which has length (1.8 m), then the uncertainty in length (ΔL) is:

a) 3mm

b) 4.5mm

c) 8.5mm

d) 6mm ✓

2) If the relation between A & B given as ($C=A-B$), if $A=5.28$ and $B=1.8$, then ($C \pm \Delta C$) is: (Using significant rules)

a) 3.48 ± 0.05

b) 3.5 ± 0.1

c) 3.5 ± 0.06 ✓

d) 3.48 ± 0.055

3) F_1, F_2, F_3 are three forces in equilibrium, if you know that ($F_{3x} = -1.32 \text{ N}$, $F_{3y} = -7.5 \text{ N}$), the direction of resultant between (F_1 & F_2) according to x-positive axis is :

a) 100

b) 80 ✓

c) 280

d) 260

4) In Newton's second law experiment, if we got this graph,

$1/Mw + mh (\text{kg}^{-1})$

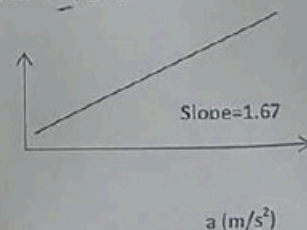
then the constant hanged mass in the experiment is:

a) 15g

b) 25g ✓

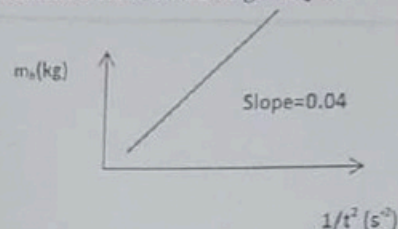
c) 60g ✓

d) 80g



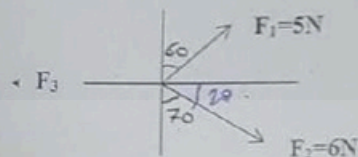
5) The graph shown below is from Newton's second law experiment, If the distance that (M_w) moved in all trials was (60 cm) and total mass (300g), then the acceleration of gravity is:

- a) 8 m/s^2 b) 7 m/s^2
 c) 9 m/s^2 d) 12 m/s^2



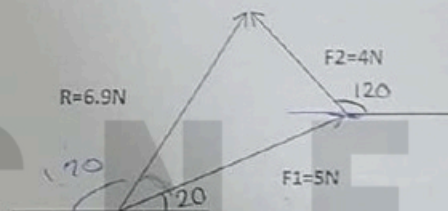
6) For equilibrium state, the magnitude of (F_3) is:

- a) 9.5N b) 9.98 N
 c) 8.45N d) 10.8N



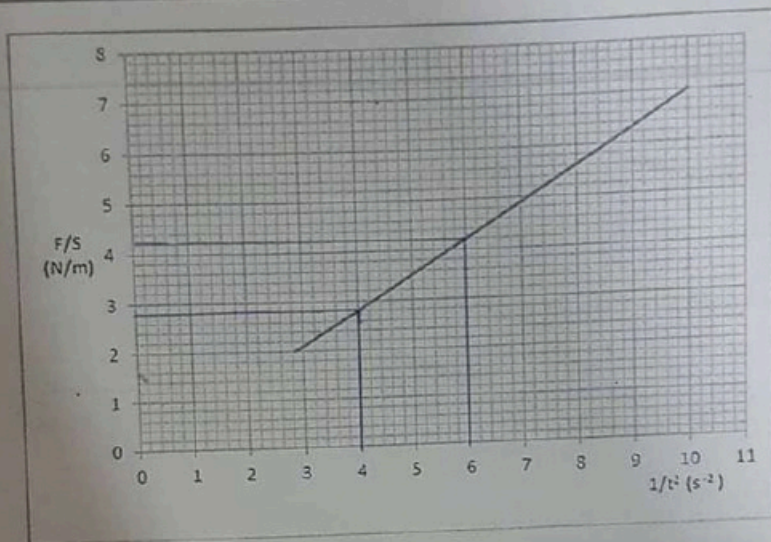
7) The direction of (R), according to x-positive axis's is:

- a) 82.8° b) 96.3°
 c) 61.8° d) 54.8°



Q3 : In Newton's second law experiment, the distance that (M_w) was moved in all trials considered to be (80cm). then use the graph to answer the following :

- (2 marks) a) The physical meaning of the slope is ($\frac{\text{total mass}}{m} \text{ (m)}$)
 (2 marks) b) The total mass is (0.2 kg)
 (2 marks) c) If ($t=0.4\text{s}$) the hanged mass (m_h) will be (2 kg)



C.N.E

COMPUTER NETWORK
ENGINEERING

Good luck



Fall 2016
Date: April 7, 2016

Midterm
Physics 111

(5-2)

Name: [REDACTED]

Instructor: [REDACTED]

Consider $g = 10 \text{ m/s}^2$

1) Circle the correct the best answer for the following questions.

Q.1 A particle of mass 5 Kg is just starting to move when a force of 30 N is applied horizontally. What is the coefficient of static friction μ_s ?

- a) 0.3 b) 1.5 c) 0.7 d) 0.5 **e) 0.6**

Q2 A student wanted to calculate the density (ρ) of Mass (M). If $M \pm \Delta M = (10 \pm 0.01)$ gram and its volume $V \pm \Delta V = (5 \pm 0.1) \text{ cm}^3$, the absolute error of the density ($\Delta \rho$) in g/cm^3 is:

- a) 0.08 **b) 0.04** c) 0.06 d) 0.02 e) none

Q.3 A Vernier caliper was used to measure the length of a cube. What is the error in length (ΔL)?

- a) 0.50 mm **b) 0.05 mm** c) 0.5 mm **d) 0.005 mm** e) 0.0005 mm

Q.4 Which of the following measurements represents the reading of the micrometer?

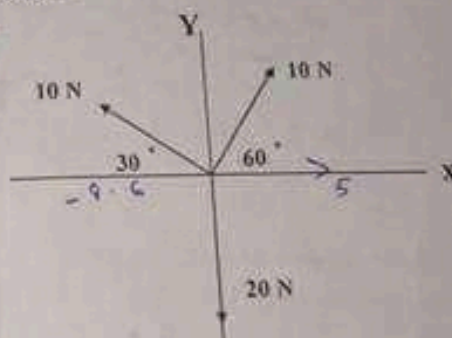
- a) 9.25 mm** b) 8.0 cm **c) 1.50 cm** d) 2 mm e) 10.1 mm

Q.5 What is the value of 0.484×0.080 to the appropriate number of significant figures?

- a) 0.04 **b) 0.03** **c) 0.039** d) 0.0387 e) 0.038

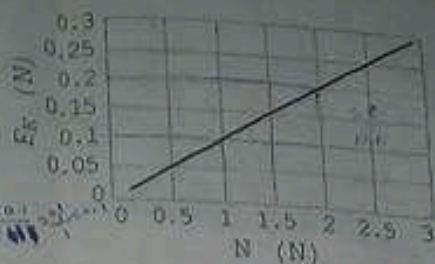
Q.6 In the Figure shown.
What is the magnitude of the resultant force (in N) along x axis?

- a) 13.66 b) 5.00 c) 8.66 **d) 3.66** e) 0



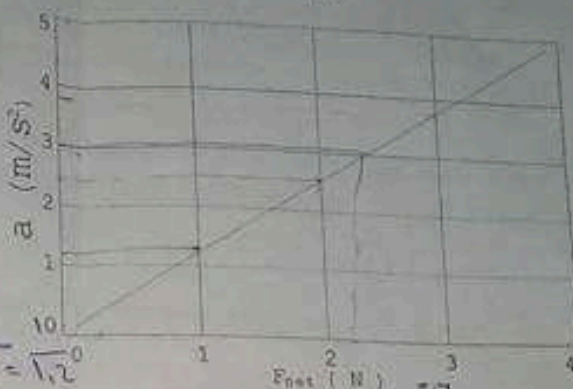
Q.7 From the graph shown. What is the coefficient of kinetic friction μ_k ?

- a) 0.1 b) 0.3 c) 0.4 d) 0.5
e) 0.2



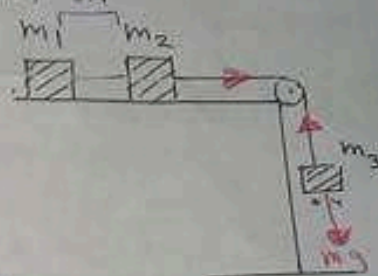
2
In the Newton's second law experiment, we add a new mass m_3 , connected to m_2 as shown in the figure. (Show your calculations and write the final answer in the box).

- a) What is the sum of the three masses.



$$\Sigma m = \text{slope} = \frac{\Delta a}{\Delta F} = \frac{2-1}{2-0} = 1.2$$

12 0.8 Kg



- b) Find F_{net} when $m_3 = 0.4$ Kg

$$m_1 + m_2 = m_3$$

$$m_1 + m_2 = 0.4$$

$$F = ma = 0.4 \times 9.8 = 3.92 \text{ N}$$

3.92 N

- c) What is the acceleration of the system when $m_3 = 0.3$ Kg?

$$F = ma$$

$$\frac{3.92}{0.3} = \frac{0.3 \times 9.8}{a}$$

$$a = 13.06$$

13.0 m/s²

$$a = 3.75 \text{ m/s}^2$$



مؤيد المختبر :

مدرس المختبر :

الاسم بالعربية :

Circle the correct answer

1. The number of significant figures in the measurement ($L = 75.00 \text{ m}$) is:

- a) 1 b) 2 c) 3 d) 4

2. The addition of the measurements (6.201 cm, 7.4 cm, 0.68 cm, and 12.0 cm) equals

- a) 26.281 cm b) 26.3 cm c) 26.28 cm d) 26.6 cm

3. The multiplication of the measurements ($3.2145 \text{ km} \times 4.23 \text{ km}$) equals

- a) 13.6 km² b) 13.597335 km² c) 13.59 km² d) 13.60 km²

4. Which of the following measurements represent the reading of the micrometer?

- a) 10.3 mm b) 8.0 cm c) 1.50 cm d) 2.32 mm

5. The reading of the micrometer is:

- a) 12.90 mm b) 10.240 mm
c) 12.40 mm d) 10.60 mm



6. The reading of the Vernier caliper is:

- a) 2.30 cm b) 2.38 cm
c) 1.48 cm d) 2.48 cm

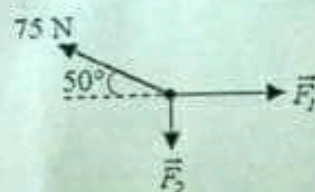


7. A cylinder has a radius of $R \pm \Delta R = (2.00 \pm 0.01) \text{ cm}$ and a length of $L \pm \Delta L = (10.0 \pm 0.1) \text{ cm}$. The error in the volume ΔV is:

- a) 0.02 cm³ b) 0.015 cm³ c) 2.51 cm³ d) 1.9 cm³

8. A stationary object has three forces acting on it, as shown in the diagram. What are the magnitude of $\vec{F}_1$?

- a) 57.5 N b) 48.2 N c) 15 N d) -20 N



لجنة الميكانيك - الإتجاه الإسلامي

Al-Balqa Applied University

Faculty of Engineering Technology

Applied Sciences Department

جامعة البلقاء التطبيقية

كلية الهندسة التكنولوجية

قسم العلوم التطبيقية

Midterm Exam

الاسم

مدرس المادة:
موقع المختبر: 2-5 الإثني

Q1: Determine the number of significant figures in the following measured values:

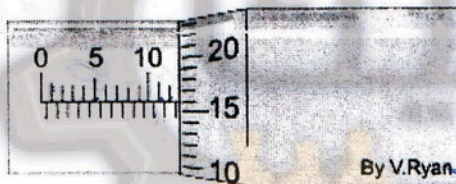
- 1- 2.3 cm 2 numbers
- 2- 3.589 s 4 numbers
- 3- 4.56×10^{-6} kg 3 numbers
- 4- 1.0003 J 5 numbers

$g = 9.8 \text{ m/s}^2$
دالة الرقم المعنوي

Q2: Carry out the following arithmetic operations :

- 1- The sum of the measured values 756, 37.2, 0.83 and 2.5 = 797
- 2- The product $0.0032 \times 356.3 = 1.1$
- 3- The division $\frac{1.05}{5.346} = 0.196$
- 4- The subtraction $0.082 - 1.02 = -0.94$

Q3- Read the scale of the micrometer, what is the error of the reading



By V.Ryan

~~125.16~~ 12.66 mm

Q4: Vector Experiment

If you have three forces, one of A ($m_1=0.200$ Kg) pointing at 30° , the second B ($m_2=0.200$ Kg) pointing at 120° , and the third C as shown in the force table. If the ring remains centered around the pin without being moved in any direction, complete the following Table

Vector	Mass, kg	Magnitude in (N)	Direction	x-component
A	0.200	2	30°	$2 \times \cos 30^\circ = 2 \times 0.86 = 1.72$
B	0.200	2	120°	1 in negative direction of x
C	0.300	3	72°	1

$\vec{A} = 1.72 \hat{i} + 1 \hat{j}$
 $\vec{B} = -1 \hat{i} + 1.72 \hat{j}$

$1 + 3 \hat{j}$
 1×0.200

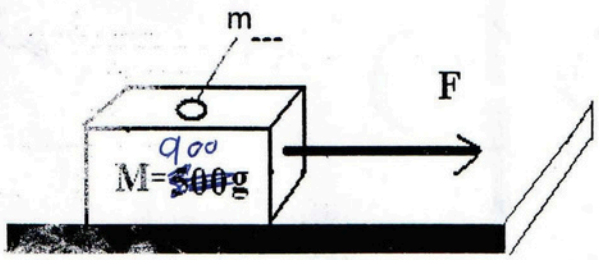
1×21.56



لجنة الميكانيك - الإتجاه الإسلامي

Q5- Friction Experiment: If the applied force F moves the block M and the mass m with constant velocity, find the kinetic coefficient of friction in each case and its average value.

$g = 9.8$



μ_k	$m(\text{kg})$	$F(\text{N})$
0.61	0.100	6.00
0.63	0.200	6.80
0.62	0.300	7.30

$F - f_k = 0$
 $f_k = F$
 $\mu_k = \frac{f}{N} = \frac{F}{(m+M)g} = \frac{6.00}{9.8} = 0.61$

$\mu_k \text{ average} = 0.62$

Q6- Measurements

Ahmad measured the radius and the mass of a sphere using the micrometer and the balance, $R = 0.86 \text{ mm}$ and $m = 4.3 \text{ g}$, $\Delta R = 0.005 \text{ mm}$, $\Delta m = 0.05 \text{ g}$

Calculate the volume ($V \pm \Delta V$) and the density ($\rho \pm \Delta \rho$) and the error in each of them.

$$V = \frac{4}{3} \pi R^3 = 2.7 \times 10^{-9} \text{ m}^3$$

$$\frac{\Delta V}{V} = \frac{3 \Delta R}{R} \Rightarrow \Delta V = \frac{2.7 \times 10^{-9} \times 3 \times 0.005 \times 10^{-3}}{0.86 \times 10^{-3}} = 0.05 \times 10^{-9}$$

$$V \pm \Delta V = (2.7 \times 10^{-9} \pm 0.1 \times 10^{-9}) \text{ m}^3$$

$$\rho = \frac{m}{V} = \frac{4.3 \times 10^{-3}}{2.7 \times 10^{-9}} = 1.6 \times 10^6 \text{ (Kg/m}^3\text{)}$$

$$\Delta \rho = \rho \left(\frac{\Delta m}{m} + \frac{\Delta V}{V} \right) = 1.6 \times 10^6 \left(\frac{0.05 \times 10^{-3}}{4.3 \times 10^{-3}} + \frac{0.05 \times 10^{-9}}{2.7 \times 10^{-9}} \right)$$

$$\Delta \rho = 0.05 \times 10^6 \therefore \rho \pm \Delta \rho = (1.6 \times 10^6 \pm 0.1 \times 10^6) \text{ Kg/m}^3$$

لجنة الميكانيك - الإتجاه الإسلامي

Al- Balqa' Applied University
Faculty of Engineering
Technology
Department of Applied Science
B



Physics 111
Midterm Exam

Time: 50 minutes

جامعة البلقاء التطبيقية
كلية للهندسة التكنولوجية
قسم العلوم التطبيقية

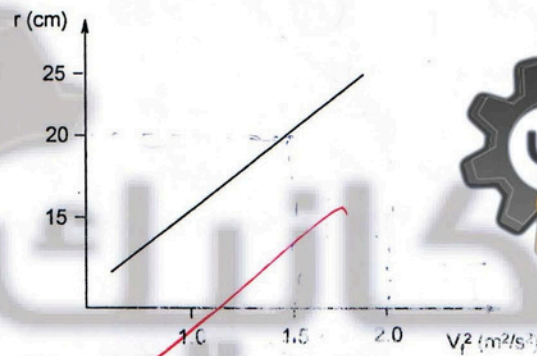
الاسم بالعربية: الأشعث الشعبة: الفيزياء المدرس: حلوه دة

Circular motion

1) A man is moved from rest around circular path with radius 60 m and he made 10 turns in 10 minute. His final velocity in (m/s) is

- a) 7.00 b) 6.28 c) 4.19 d) 3.50

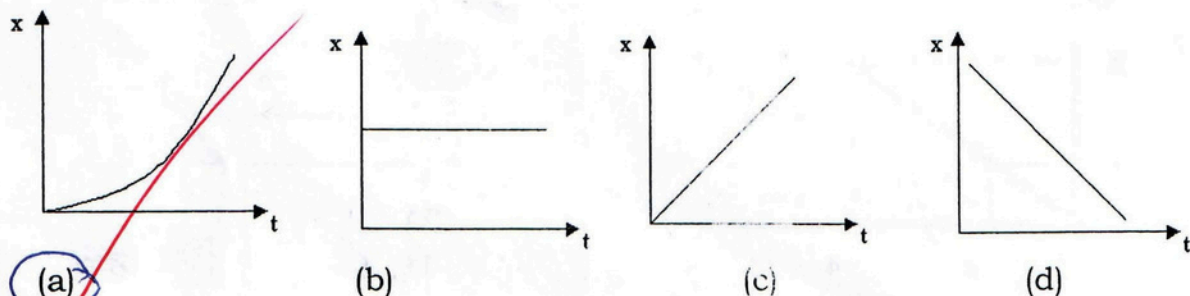
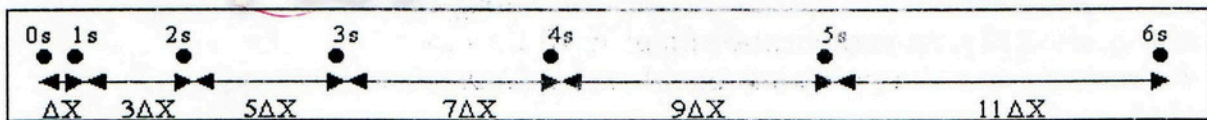
2) From the graph, the experimental centripetal acceleration is



- a) 0.1 m/s² b) 10 m/s² c) 15 m/s² d) 0.15 m/s²

Kinematics of rectilinear motion

3) For the ticker tape shown below, the correct graph representing position versus time is.



(a)

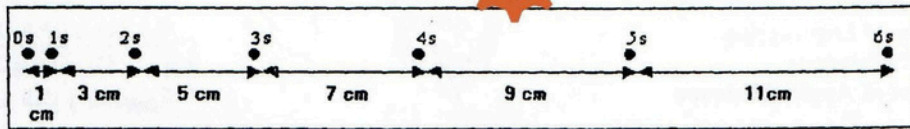
(b)

(c)

(d)



لجنة الميكانيك - الإتجاه الإسلامي



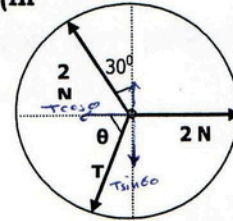
4) For the ticker tap shown above, the average velocity (in **cm/s**) for the period $t = 1s$ to $t = 4s$ is:

- (a) 3 (b) 2.5 (c) 5 (d) 3.5

Vector: Force table

5) For the experimental setup shown. If the body located, at the center, and its at equilibrium. The values of **T (in N)**, and **θ (in degree)** are respectively:

- a) 2.64, 71.9 (b) 3.36, 68.9 (c) 1.98, 59.83 (d) 3.36, 79.1 (e) 2.64, 79.1

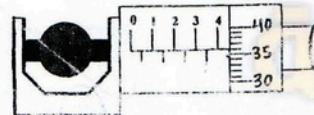


Measurments and uncertainties

6) A micrometer is used to measure the diameter of a sphere as shown in the figure.

The volume of the sphere in (**mm³**) is:

- a) 37.7 (b) 347 (c) 43.4 (d) 302



7) A Vernier caliper was used to measure the length of a cube, the error in length ΔL is

- a) 1mm (b) 0.5mm (c) 0.05mm (d) 0.1mm

Experimental setup shown in Fig.1, when the masses $M_a + M_b$ were kept constant and equal to 210 g, the graph between $M_b g$ the hanging mass times gravity acceleration, and a , is the acceleration is shown in Fig.2.

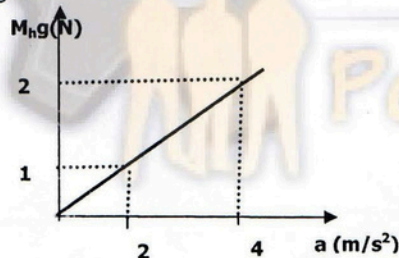
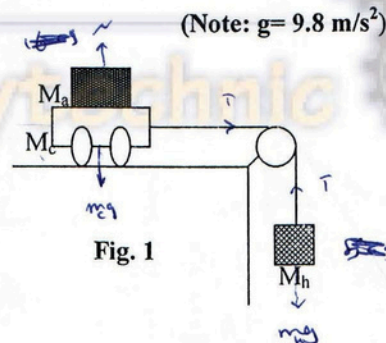


Fig.2



Q8) The mass of the cart M_c (in gm) is :

Q9) If the added mass is $M_a = 120$ g, the magnitude of the acceleration for this system (in **m/s²**) is:

$$\begin{aligned} F &= ma \\ F &= (M_c + M_a) a \\ m_h g &= \frac{F}{a} = \frac{2}{4} = 0.5 \text{ g} \\ 2 &= (M_c + 209.5) 4 \\ 0.5 &= M_c + 209.5 \\ M_c &= 209 \end{aligned}$$

$$\begin{aligned} F &= ma \\ a &= \frac{F}{m} \\ a &= \frac{0.5}{4/8} = 1.1962 \times 10^{-3} \text{ m/s}^2 \end{aligned}$$

لجنة الميكانيك - الإتجاه الإسلامي

Al- Balqa' Applied University
Faculty of Engineering
Technology
Department of Applied Science



Physics 111
Midterm Exam
Time: 50 minutes

18.5
20

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم التطبيقية

موعد المختبر : الخميس (٥-٢)

مدرس المختبر: الدكتور زياد عليما

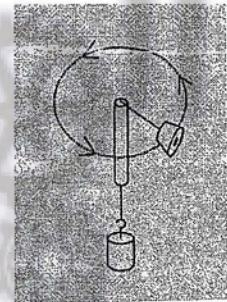
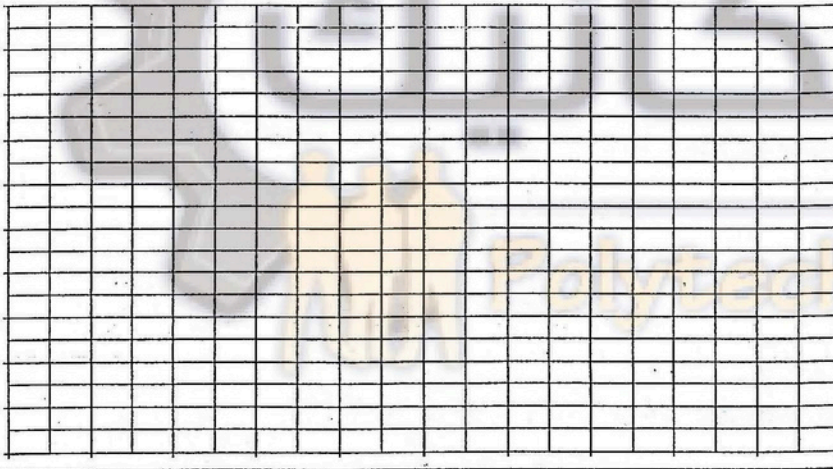
الاسم بالعربية:

Q1: Uniform Circular Motion

a) In the circular motion experiment, a student studied the relation between the centripetal force and the square of the speed (v^2) for rotational body (m) by making the radius ($r=0.45$ m).

M (kg)	Period T (s)	F_c (N)	v (m/s)	v^2 (m/s) ²
0.02	1.15			
0.04	0.82			
0.06	0.67			

- Complete the table?
- Plot a suitable graph between F_c and v^2
- By using the slope of the graph, find the mass (m) of the rotational body.



Q.2) A ball of mass 0.5 kg is attached to the end of a string 1.5 m long and whirled in a horizontal plane at constant speed. If the cord can withstand a maximum tension of 50 N, what is the maximum speed of the ball before the string breaks?

38.7 m/s

Q.2: Friction Force:

Q.1) For the inclined surface, if $\mu_k = \mu_s = 0.3$; if the mass slides on the surface with constant velocity. Find the angle of the inclined surface?

$$M_s = \tan \theta$$

$$0.3 = \tan \theta$$

$$\theta = \tan^{-1} 0.3$$

$$\theta = 16.7^\circ$$



لجنة الميكانيك - الإتجاه الإسلامي

Q.2) the force $F=4N$, is applied to move the mass $M=500$ gram, on friction surface with constant velocity, find the kinetic coefficient of friction between the mass and the surface?

[The gravity acceleration = 9.81 m/s^2]

$\mu_k = \frac{F_f}{F_N} = \frac{F}{mg} = \frac{4}{0.5 \times 9.81} = 0.815$

Q3: Vectors $4 = \mu_k \times 500 \times 10^{-3} \times 9.81$

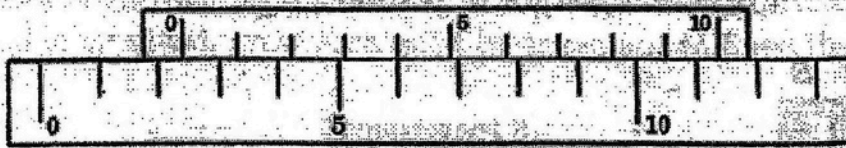
$\mu_k = 0.815$

Find the resultant force of the following two forces: $F_1 = 1.96 \text{ N}$ (weight of 200 gram) at 60° from positive x axis. $F_2 = 2.94 \text{ N}$ (weight of 300 gram) at 140° from the positive x axis.

The resultant force = 3.81 N, with direction = 109.5 degree

Q4: Measurements

a) Read the scale of the Vernier:



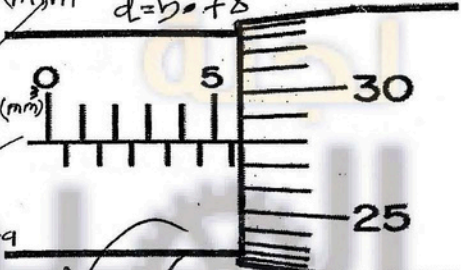
b) In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was $M \pm \Delta M = (2.1 \pm 0.05) \text{ g}$. calculate:

a) The Radius of the sphere, $R \pm \Delta R = 2.78 \pm 0.005 \text{ (mm)}$ $d = 5.78$

b) The volume of the sphere, $V \pm \Delta V = 90.90 \pm 0.0054 \text{ (mm}^3\text{)}$

$\frac{4}{3} \pi r^3$
 $\frac{\Delta V}{V} = 3 \times \frac{\Delta R}{R} = 3 \times \frac{0.005}{2.78}$

c) The density of the sphere, $\rho \pm \Delta \rho = 0.023 \pm 0.00059 \text{ (g/mm}^3\text{)}$



Q5: Significant Figures and Errors

a) Express the results of the following calculations to the appropriate number of significant figures?

1) $7.939 + 6.26 + 11.1 = 25.3$ 2) $(27.2 \times 15.63)/1.846 = 230$

b) Rounding the following numbers:

1) (0.07284) to 3 significant digits: 0.0728 2) (10.5660) to 4 significant digits: 10.57
3) (726.835) to 3 significant digits: 727

c) Determine the number of significant figures in the following numbers:

1) $005.00 \times 10^{-4} \text{ cm}$ 3 2.) 0.0350 gram 3
3.) 1.0003 Joule 5 4.) 50000 N 1

d) If $X \pm \Delta X = (4.00 \pm 0.02)$ and $Z = \sqrt{X}$

Calculate $Z \pm \Delta Z = 2.00 \pm 0.005$

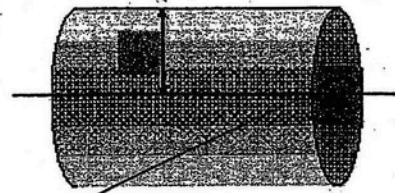
e) Experimental value of gravity is found to be 10.07 m/s^2 . Accepted value of "g" is 9.81 m/s^2 . Calculate the percent error?

$0.027 \times 100\% = 2.7\%$



6.

A long nonconducting cylinder of radius (R) has a positive charge of a uniform density (ρ) distributed throughout its volume. The magnitude of the electric field at distance r from the cylinder axis, where $r < R$ is given by:



- a) $\frac{\rho r}{4\epsilon_0}$ b) $\frac{\rho R^2}{2\epsilon_0 r}$ c) $\frac{\rho r^2}{2\epsilon_0}$ d) $\frac{\rho r}{2\epsilon_0}$ e) $\frac{\rho R^3}{2\epsilon_0 r^2}$

7. A uniform linear charge density of 4.0 nC/m is distributed along the entire x axis. Consider a spherical (radius = 5.0 cm) surface centered on the origin. The electric flux (in $\text{N m}^2/\text{C}$) through this surface is equal to:

- a) 45 b) 68 c) 62 d) 79 e) 23

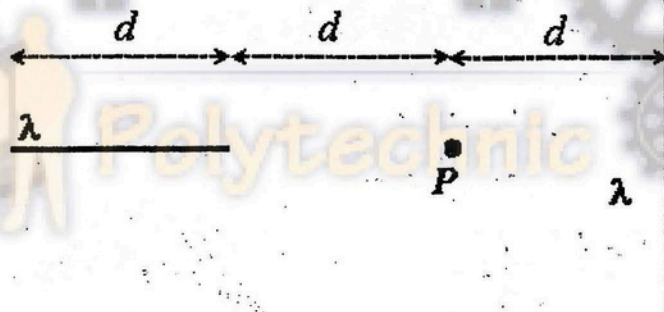
8. Positive charge Q is distributed uniformly throughout an insulating sphere of radius R centered at the origin. A particle with positive charge Q is placed at $x = 2R$ on the x axis. The magnitude of the electric field at $x = R/2$ on the x axis is:

- a) $\frac{17Q}{72\pi\epsilon_0 R^2}$ b) $\frac{Q}{8\pi\epsilon_0 R^2}$ c) $\frac{Q}{72\pi\epsilon_0 R^2}$ d) $\frac{Q}{4\pi\epsilon_0 R^2}$

9. A charge of $+80 \mu\text{C}$ is placed on the x -axis at $x = 0$. A second charge of $-50 \mu\text{C}$ is placed on the x -axis at $x = 50 \text{ cm}$. the magnitude of the electrostatic force on a third charge of $4 \mu\text{C}$ placed on the x -axis at $x = 30 \text{ cm}$ is equal to:

- a) 45 N b) 77 N c) 25 N d) 13 N e) 39 N

10. In the figure shown two rods with uniform linear charge density (λ), the vertical rod is infinite and the horizontal rod is finite with a length (d), the magnitude of the electric field resulting from both rods at the point (P) is :



- a) $3k\lambda/2d$ b) $2k\lambda/d$ c) $5k\lambda/2d$ d) $k\lambda/2d$

لا شيء اكل
لا يضاف

$$\frac{2\lambda ke}{2d} + ke\lambda \left(\frac{1}{2d}\right)$$

$$2 \times \frac{2\lambda ke}{2d} + \frac{ke\lambda}{2d}$$

$$\frac{4\lambda ke + ke\lambda}{2d} = \frac{5ke\lambda}{2d}$$

لجنة الميكانيك - الإتجاه الإسلامي

Al- Balqa' Applied University
Faculty of Engineering
Technology
Department of Applied Science



Physics 111
Midterm Exam
Time: 50 minutes (A)

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم التطبيقية

مدرس المختبر: د. محمود حاتم

موعد المختبر: ٥-٢

الاسم بالعربية:

Q1: Significant Figures and Errors

a) Express the results of the following calculations to the appropriate number of significant figures?

1) $0.424 \times 0.090 = 0.038$

2) $\frac{1.36 \times 0.5}{2.6} = 0.3$

4) $37.76 + 3.907 + 226.4 = 268.1$

3) $3.76 + 14.83 + 2.1 = 20.7$

b) If $X \pm \Delta X = (4.00 \pm 0.02)$ and $Z = \sqrt{X}$

Calculate $Z \pm \Delta Z = 2.00 \pm 0.005$

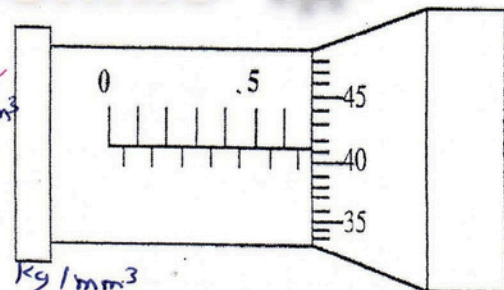
Q2: Measurements

In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was $M \pm \Delta M = (2.1 \pm 0.05)$ g. calculate:

a) The Radius of the sphere, $R \pm \Delta R = \left(\frac{6.91 \pm 0.005}{2} \right) \text{ mm}$

b) The volume of the sphere, $V \pm \Delta V = 1382.04 \pm 0.6 \text{ mm}^3$

c) The density of the sphere, $\rho \pm \Delta \rho = \frac{2.1 \pm 0.05}{1.319 \pm 0.036} = 1.59 \pm 0.036 \text{ kg/mm}^3$



Q3: Vectors

Given three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$, if $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$, complete the following table:

Force	Mass (kg)	Force (N)	Direction	x-component	y-component
$\vec{F}_1$	0.15 kg	1.5 N	0°	$F_1 \cos 0 = 1.5$	$F_1 \sin 0 = 0$
$\vec{F}_2$	0.26 kg	2.6 N	90°	0	2.6
$\vec{F}_3$	0.300 kg	$0.3 \times 10 = 3 \text{ N}$	240°	-1.5	-2.6

Q4: Newton's Second Law

In the Newton's second law experiment, if the traveled distance (s) during a time (t) is (s = 0.8m).

a) Complete the following table:

Hanging mass m_h (kg)	Time t (s)	Force (N)	Acceleration a (m/s ²)	Total mass (kg)
0.02	1.40	0.2	0.8	0.245
0.04	0.98	0.4	1.66	0.24
0.06	0.8	0.6	2.5	0.24

b) Write your conclusion?

when the mass increase the force increase too, ~~but~~ time decrease.
 $a = \frac{2s}{t^2}$, $F = m_h g$

Q5: Uniform Circular Motion

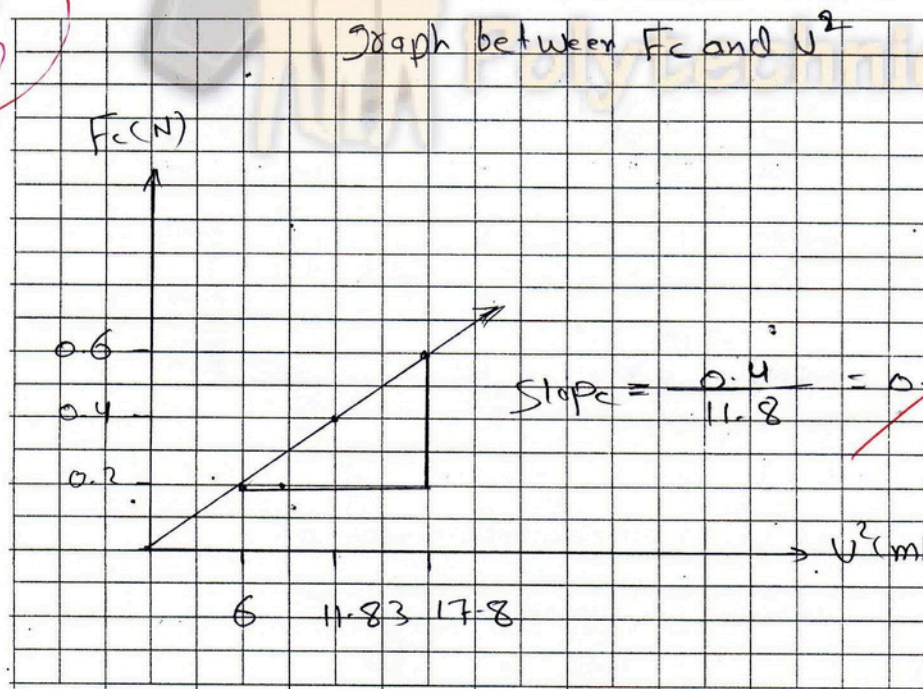
In the circular motion experiment, a student studied the relation between the centripetal force and the square of the speed (v^2) for rotational body (m) by making the radius (r=0.45 m).

M (kg)	Period T (s)	F_c (N)	v (m/s)	v^2 (m/s) ²
0.02	1.15	0.2	2.45	6
0.04	0.82	0.4	3.448	11.83
0.06	0.67	0.6	4.219	17.8

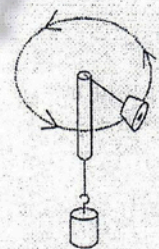
a) Complete the table?

b) Plot a suitable graph between F_c and v^2

c) By using the slope of the graph, find the mass (m) of the rotational body.



$$\text{Slope} = \frac{0.4}{11.8} = 0.033 \text{ N/m/s}^2$$



$\text{Slope} = \frac{4\pi^2 r}{m T^2}$

$$m = \frac{\text{Slope}}{4\pi^2 r}$$

$$m = 0.03 \text{ kg}$$



Name (بالعربي): محمد ابراهيم الكيا... Reg. Number: ٣١٥٠٧١٧٨٠٣٢ Date: 30/06/2016

Class Number: Instructor Name: Class Date & Time:

***MEASUREMENT AND UNCERTAINTIES:**

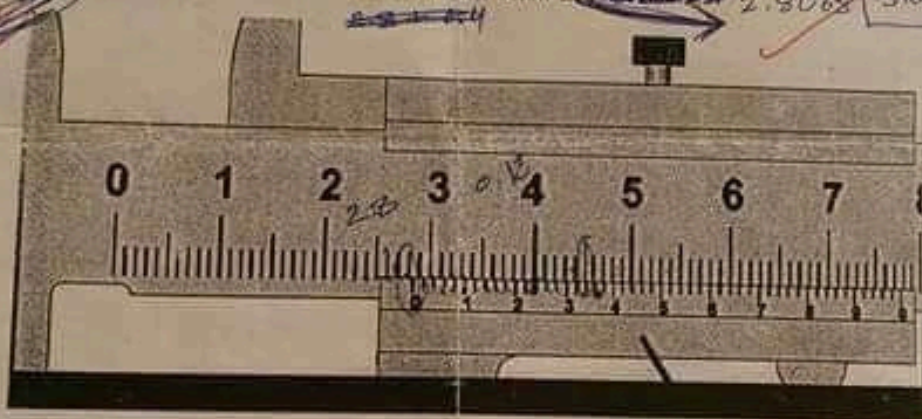
1) Fill in the blanks with the correct word or sentence: (6 marks)

(a) The error in computing the volume of cylinder ΔV Using the formula $V = \pi r^2 L$ is given by

(b) The standard error of measurement (SEM) for the following lengths reading (4.5, 4.4, 4.3, 4.6) is: 0.124016

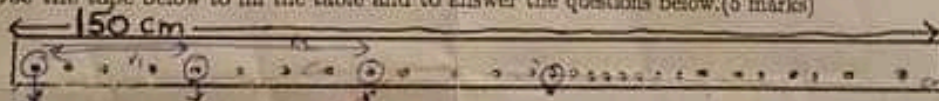
(c) If all of the scale lines of the main scale and vernier scale are all in alignment what is the reading of the vernier scale: the angle between them is zero and

The reading of the vernier calibre shown below is: 2.8068



***KINEMATIC OF RECTILINEAR MOTION:**

2) Use the tape below to fill the table and to answer the questions below. (5 marks)



Time (s)	Position (cm)	Average Velocity (cm/s)	Average Acceleration (cm/s ²)
0.1	25		
0.15		270	
0.2	52		300
0.25		300	
0.3	82		

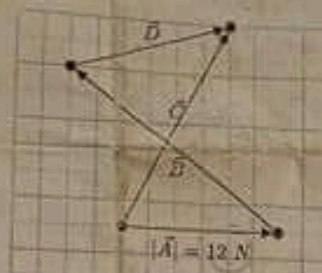
(a) The velocity during the time interval $t \in [0.3, 0.5]$ sec is: (Decreasing, Increasing, Constant).

(b) The acceleration during the time interval $t \in [0.2, 0.5]$ sec is: (Negative, Zero, Positive).

50 dot $\rightarrow$ is
5 dot $\rightarrow$ is
 $\Delta t = 0.1$ s

*VECTORS:

- 3) The figure below show the vector polygon method (graphical method) applied to find the resultant vector. Use the figure to answer the following (5 marks)



- (a) Which of the above vectors represent the resultant vector: C
- (b) The two components of the resultant vector are: $R_x = 12 \text{ N}$, $R_y = 13.04 \text{ N}$
- (c) If instead we use the force table, what would the the balance force angle be 62°

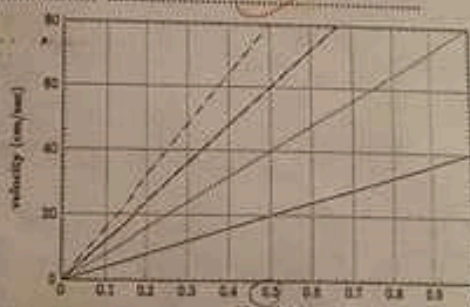
*FORCE AND MOTION:

- 4) Answer the following questions concerning this force and motion experiment (5 marks)

- (a) What is the objectives of this experiment: find the magnitude of the (v, a) and find the total and m_c and m_a

- (b) Use the adjacent figure to fill the following table:

Added mass m_c (g)	Acceleration a (cm/s^2)
100	4.0
200	8.0
300	12.0
400	16.0



- (c) Plot a graph between the added mass (y-axis) and the acceleration (x-axis).
- (d) The slope of the line of your graph is: total (unit = g) it represents: 2.5 g

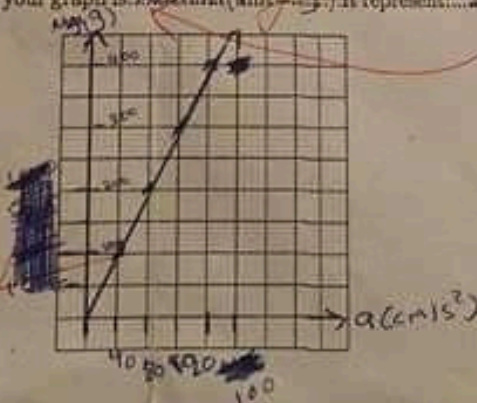
$$\text{Slope} = \frac{200 - 100}{8.0 - 4.0}$$

$$= \frac{100}{4.0} = 2.5$$

$$m_a + m_h + m_c = m_{\text{total}}$$

$$0.5 = m_{\text{total}} - 0.5$$

$$m_c = 2.5 - 0.5 = 2$$



GOOD LUCK



Circle the correct answer

1. The number of significant figures in the measurement ($L = 75.00 \text{ m}$) is:

a) 1

b) 2

c) 3

d) 4

2. The addition of the measurements (6.201 cm, 7.4 cm, 0.68 cm, and 12.0 cm) equals

a) 26.281 cm

b) 26.3 cm

c) 26.28 cm

d) 26.0 cm

3. The multiplication of the measurements ($3.2145 \text{ km} \times 4.23 \text{ km}$) equals

a) 13.6 km^2

b) 13.597335 km^2

c) 13.59 km^2

d) 13.60 km^2

4. Which of the following measurements represent the reading of the micrometer?

a) 10.3 mm

b) 8.0 cm

c) 1.50 cm

d) 2.32 mm

5. The reading of the micrometer is:

a) 12.90 mm

b) 10.240 mm

c) 12.40 mm

d) 10.60 mm



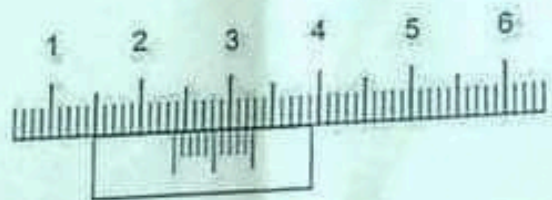
6. The reading of the Vernier caliper is:

a) 2.30 cm

b) 2.38 cm

c) 1.48 cm

d) 2.48 cm



7. A cylinder has a radius of $R \pm \Delta R = (2.00 \pm 0.01) \text{ cm}$ and a length of $L \pm \Delta L = (10.0 \pm 0.1) \text{ cm}$. The error in the volume ΔV is:

a) 0.02 cm^3

b) 0.015 cm^3

c) 2.51 cm^3

d) 1.9 cm^3

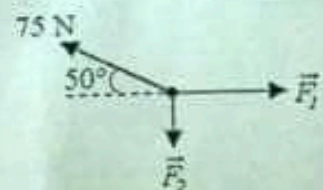
8. A stationary object has three forces acting on it, as shown in the diagram. What are the magnitude of $\vec{F}_1$?

a) 57.5 N

b) 48.2 N

c) 15 N

d) -20 N





Name:

Consider $g = 10 \text{ m/s}^2$ %

Instructor:

1) Circle the correct the best answer for the following questions.

Q.1 A particle of mass 5 Kg is just starting to move when a force of 30 N is applied horizontally. What is the coefficient of static friction μ_s ?
a) 0.3 b) 1.5 c) 0.7 d) 0.5 **e) 0.6**

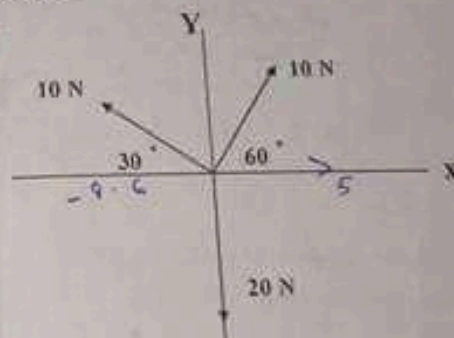
Q2 A student wanted to calculate the density (ρ) of Mass (M). If $M \pm \Delta M = (10 \pm 0.01)$ gram and its volume $V \pm \Delta V = (5 \pm 0.1) \text{ cm}^3$, the absolute error of the density ($\Delta \rho$) in g/cm^3 is:
a) 0.08 **b) 0.04** c) 0.06 d) 0.02 e) none

Q.3 A Vernier caliper was used to measure the length of a cube. What is the error in length (ΔL)?
a) 0.50 mm **b) 0.05 mm** c) 0.5 mm **d) 0.005 mm** e) 0.0005 mm

Q.4 Which of the following measurements represents the reading of the micrometer?
a) **9.25 mm** b) 8.0 cm **c) 1.50 cm** d) 2 mm e) 10.1 mm

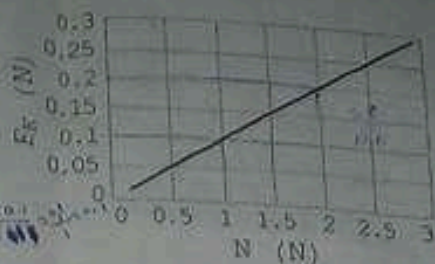
Q.5 What is the value of 0.484×0.080 to the appropriate number of significant figures?
a) 0.04 **b) 0.03** **c) 0.039** d) 0.0387 e) 0.038

Q.6 In the Figure shown.
What is the magnitude of the resultant force (in N) along x axis?
a) 13.66 b) 5.00 c) 8.66 **d) 3.66** e) 0



Q.7 From the graph shown. What is the coefficient of kinetic friction μ_k ?

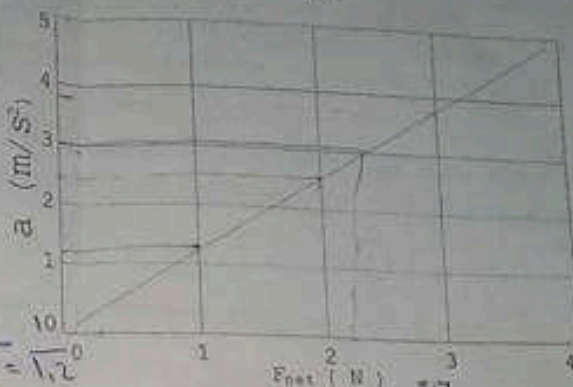
- a) 0.1 b) 0.3 c) 0.4 d) 0.5
e) 0.2



2

In the Newton's second law experiment, we add a new mass m_3 , connected to m_2 as shown in the figure. (Show your calculations and write the final answer in the box).

- a) What is the sum of the three masses.



$$\Sigma m = \text{slope} = \frac{\Delta a}{\Delta F} = \frac{2-0}{2.6-0} = 1.2$$

12 0.8 Kg

- b) Find F_{net} when $m_3 = 0.4 \text{ Kg}$

$$m_1 + m_2 = m_3$$

$$m_1 + m_2 = 0.4$$

$$F = ma = 0.4 \times 9.8 = 3.92 \text{ N}$$

3.92 N

- c) What is the acceleration of the system when $m_3 = 0.3 \text{ Kg}$?

$$F = ma$$

$$3.92 = 0.3 \times a$$

$$a = \frac{3.92}{0.3} = 13.06$$

13.06 m/s^2

$$a = 3.75 \text{ m/s}^2$$

